Problem Set 1

Starter

1 (E)**.** Let A and B denote square matrices with real entries. For each of the following assertions, find either a proof or a counterexample.

(i) $A^2 - B^2 = (A - B)(A + B)$.

(ii) If $AB = 0$ then $A = 0$ or $B = 0$.

(iii) If $AB = 0$ then A and B cannot both be invertible.

(iv) If A and B are invertible then $A + B$ is invertible.

2 (E)**.** Let $b \in \mathbb{F}$. Show that $\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$ is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

3 (L-M)**.** Do there exist matrices $A, B \in M_n(\mathbb{R})$ such that $AB - BA = I_n$?

4 (L-M)**.** Let V be a vector space over $\mathbb{R}$.

(i) Show that the set $V^{\mathbb{C}} = V \times V$ of ordered pairs, with componentwise addition

$$(u, v) + (x, y) = (u + x, v + y)$$

and scalar multiplication over $\mathbb{C}$ defined by

$$(a + bi)(u, v) = (au - bv, av + bu), \quad a, b \in \mathbb{R}$$

is a vector space over $\mathbb{C}$. Here $V^{\mathbb{C}}$ is called the *complexification* of V and we can write $ti = (0, t)$ for all $t \in V$.

(ii) Let U be a subspace of $V^{\mathbb{C}}$. Prove that there is a subspace S of V for which

$$U = S^{\mathbb{C}} = \{s + ti : s, t \in S\}$$

if and only if U is closed under complex conjugation $\chi : V^{\mathbb{C}} \to V^{\mathbb{C}}$ defined by $\chi(u + iv) = u - iv$.

Advanced

5 (L-M)**.** Let T, U and W be subspaces of V.

(i) Give explicit counter-examples to the following statements:

(a) $T + (U \cap W) = (T + U) \cap (T + W)$;

(b) $(T + U) \cap W = (T \cap W) + (U \cap W)$.

(ii) Show that each of the equalities in (i) can be replaced by a valid inclusion of one side in the other.

6 (M)**.** Let W be a subspace in V over $\mathbb{F}$ and let $\mathbf{v} \in V$. The set $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} \mid \mathbf{w} \in W\}$ is called a *coset* or a *translate* of W.

(a) Show that two cosets of the same subspace W either coincide or are disjoint.

(b) Prove that if two cosets of two distinct subspaces are not disjoint, then their intersection is a coset of some subspace.

(c) Show that a nonempty subset $S \subseteq V$ is a coset of some subspace if and only if $x + \lambda(y - z) \in S$ for all $x, y, z \in S$ and $\lambda \in \mathbb{F}$.

7 (M)**.** Let V be a vector space and let $\mathcal{F} = \{S_i \mid i \in I\}$ be a family of distinct subspaces of V. Show that the following are equivalent:

(i) For each $i \in I$,
$$S_i \cap \left(\sum_{j \neq i} S_j \right) = \{0\}.$$

(ii) The zero vector 0 cannot be written as a sum of nonzero vectors from distinct subspaces of $\mathcal{F}$.

(iii) Every nonzero $v \in \sum_{i \in I} S_i$ has a unique, except for order of terms, expression as a sum
$$v = s_1 + \cdots + s_n$$
of nonzero vectors from distinct subspaces in $\mathcal{F}$.

8 (H)**.** Show that a nontrivial vector space over an infinite field is not the union of a finite number of proper subspaces.